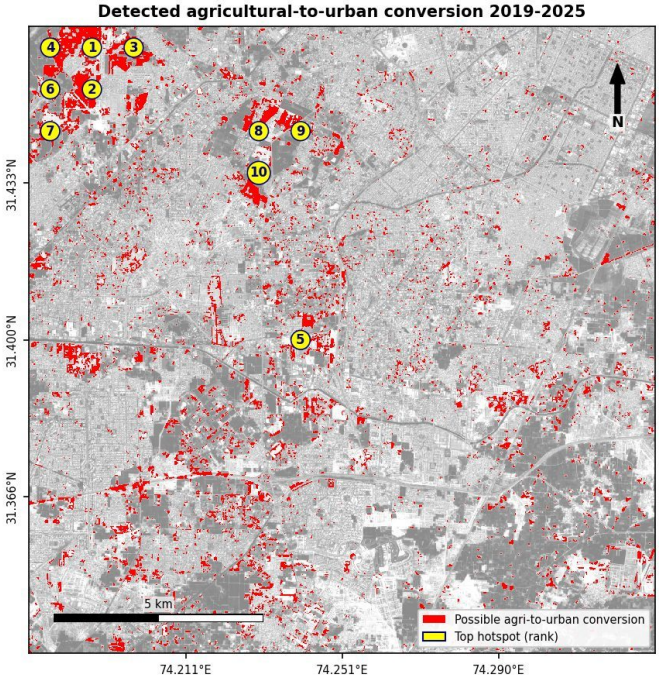
Punjab GeoGuardian

AI-Based Monitoring of Urban Expansion and Farmland Loss

Final Report - Lahore peri-urban expansion zone near Raiwind Road - 2019 vs 2025



Source: Copernicus Sentinel-2 L2A (ESA), processed in Google Earth Engine. Author: Hamza Nadeem. Estimated change - learning prototype.

Item	Detail
Prepared by	Hamza Nadeem
Submitted to	Omar Farooq - Associate AI/ML Engineer
Task	AIRI Team PITB - AI Internship Task 3 (Individual)
Area of Interest	Lahore peri-urban expansion zone near Raiwind Road
Years compared	2019 and 2025
Satellite data	Copernicus Sentinel-2 Level-2A (10 m)
Report date	11 November 2026

1. Executive Summary

- Within the 225.0 km² Area of Interest, an estimated 12.31 km² (1231 ha) was classified as vegetation/agriculture in 2019 and as built-up in 2025. This is a possible agricultural-to-urban conversion equal to 5.47% of the AOI and 14.76% of the 2019 vegetation area.
- Mapped built-up area changed from 112.82 km² in 2019 to 118.42 km² in 2025 (net +5.60 km², +5.0%).
- Mapped vegetation/agriculture changed from 83.39 km² to 73.09 km² (net -10.30 km²).
- Validation on 60 checked points: 29 correct, 31 incorrect, approximate accuracy 48.3% (area-weighted 56.6%).
- After correcting for the observed map errors, the conversion area is estimated at 33.70 km² (95% CI 10.92-56.47 km²).
- 3705 separate conversion patches were detected; 185 are 1 ha or larger.
- The largest contiguous patch covers about 67.1 ha.
- The 10 largest patches together cover 221.9 ha (18% of all detected conversion).

Important: This is a learning prototype, not an official government monitoring system. All results are *estimated* or *possible* changes detected from 10 m satellite data. They are not a finding about any specific housing society, development or landowner (see Section 11).

2. Problem Statement and Objectives

Punjab is urbanising quickly. Around large cities such as Lahore, agricultural land, green zones and peri-urban land are being converted into housing, roads and commercial areas. Monitoring this by hand is slow and inconsistent. Punjab GeoGuardian is a GeoAI prototype that compares Sentinel-2 imagery from two years, classifies land cover, detects where vegetation/agriculture became built-up, measures the area and presents the results on maps and a dashboard.

The project answers four questions:

1. How much land has changed from vegetation/agriculture to built-up area?
2. Where are the major change hotspots?
3. Can the results be visualized on a map?
4. Can the project be extended for district-level monitoring?

GeoAI workflow followed:

```
Select AOI → Load Sentinel-2 → Cloud-free median composite → NDVI and NDBI → Classify land cover → Compare old vs recent year → Detect changed pixels → Calculate area → Visualise on maps → Validate and analyse errors → Report and recommendations
```

3. Area of Interest and Data

Lahore peri-urban expansion zone near Raiwind Road. South-western peri-urban belt of Lahore along the Raiwind Road corridor, south of Thokar Niaz Baig. The area mixes established housing, new housing schemes, roads, brick kilns and wheat/rice farmland, which makes it a typical example of Lahore's outward growth onto agricultural land.

The AOI is a 15 km × 15 km rectangle centred at 31.4000°N, 74.2500°E, analysed on a 10 m grid in EPSG:32643 (UTM), total area 225.00 km². Reasons for choosing it: it is a well-known peri-urban growth corridor (the PDF's own example), it mixes farmland with new development, and its size is manageable for a two-week project while being large enough for meaningful statistics.

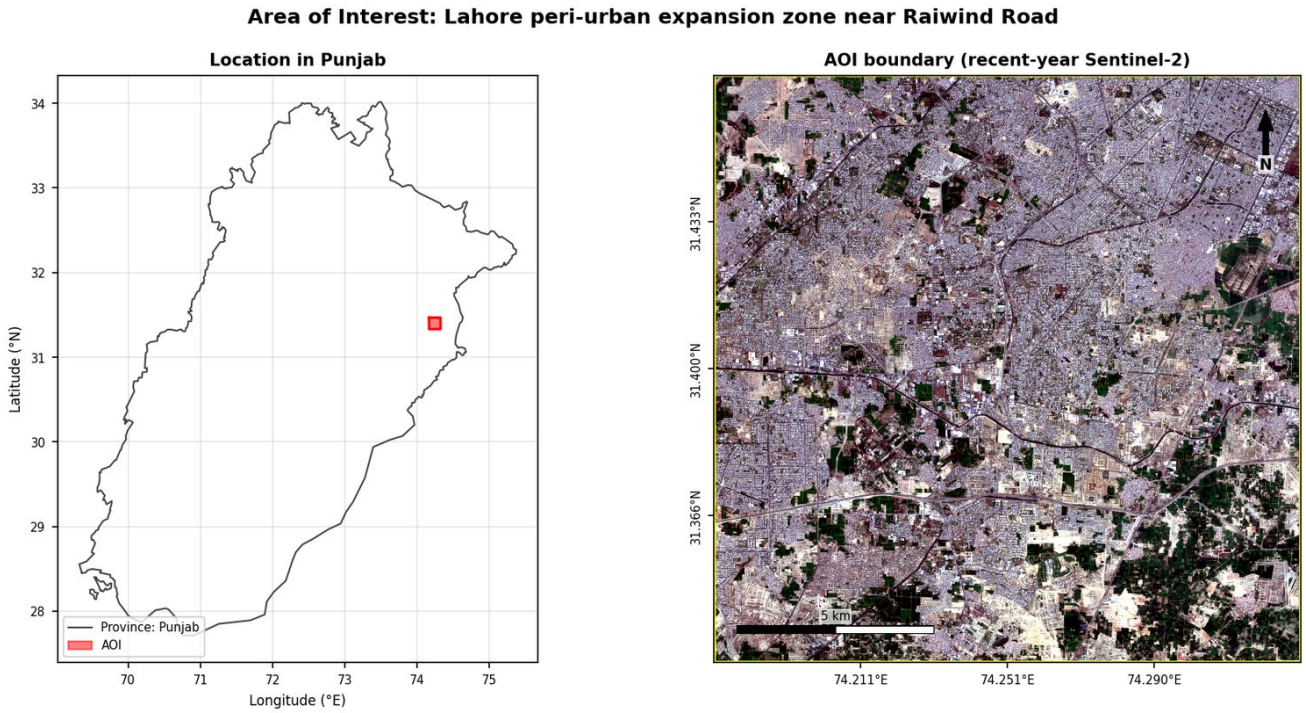


Figure 1 - AOI location and boundary.

Image selection

Year	Date range	Image dates	Scenes	MGRS tiles	Window widened
2019	2019-02-01 to 2019-03-31	11	11	43RDQ	0 d
2022	2022-02-01 to 2022-03-31	18	19	43RDQ	0 d
2025	2025-02-01 to 2025-03-31	13	14	43RDQ	0 d

- Satellite: Copernicus Sentinel-2 MSI Level-2A surface reflectance (COPERNICUS/S2_SR_HARMONIZED). The *harmonized* version removes the +1000 offset ESA added in processing baseline 04.00 (2022), so the 2019 and 2025 values are comparable.
- Same season each year (02-01 to 03-31, MM-DD): rabi wheat is green, cloud cover is low and winter smog has mostly cleared. Comparing the same crop stage reduces false change.
- Cloud filter: scenes with CLOUDY_PIXEL_PERCENTAGE < 20%, then pixel-level masking with Google Cloud Score+ (cs_cdf ≥ 0.6) and the Scene Classification Layer (clouds, shadows, cirrus, saturated pixels).
- Median composite of all clear observations, which removes leftover haze and noise; clipped to the AOI.
- Bands: Blue (B2), Green (B3), Red (B4), NIR (B8) at 10 m; SWIR1 (B11) and SWIR2 (B12) at 20 m, resampled to 10 m.

4. Remote-Sensing Concepts

NDVI - Normalized Difference Vegetation Index

NDVI = (NIR − Red) / (NIR + Red) [Sentinel-2: (B8 − B4) / (B8 + B4)]

Healthy plants reflect strongly in near-infrared and absorb red light for photosynthesis, so vegetation has a high NDVI (about 0.4-0.9). Bare soil and built surfaces are near 0.1-0.2, and water is negative. **High NDVI usually indicates vegetation or agricultural land.**

NDBI - Normalized Difference Built-up Index

NDBI = (SWIR − NIR) / (SWIR + NIR) [Sentinel-2: (B11 − B8) / (B11 + B8)]

Concrete, asphalt and roofs reflect more short-wave infrared than near-infrared, so **high NDBI often indicates built-up or urban surfaces**. Dry bare soil also has a high SWIR response, which is why NDBI on its own confuses fallow fields with built-up land. That is one reason this project combines NDBI with NDVI, all spectral bands and a machine-learning model.

MNDWI (supporting index)

MNDWI = (Green − SWIR) / (Green + SWIR) [(B3 − B11) / (B3 + B11)]

Used to separate open water and wet soil (part of the 'Other' class).

Change detection

Change detection compares data for the same place at two times. This project uses **post-classification comparison**: both years are classified with the same model, and a pixel that is Vegetation/Agriculture in 2019 and Built-up in 2025 is marked as **Agricultural-to-Urban Conversion**. To avoid counting small classification differences as change, the pixel must also show a real spectral change (NDVI drop ≥ 0.1 or NDBI rise ≥ 0.05). Isolated patches smaller than 5 pixels (500 m²) are removed.

Example: 2019: Vegetation → 2025: Built-up ⇒ Agricultural / green land converted into urban land

5. Methodology

5.1 Land classes

Class ID	Class Name	Description
0	Vegetation / Agriculture	Crops, fields, green land, trees
1	Built-up / Urban	Buildings, roads, housing societies, developed land
2	Other	Bare land, water, mixed land, unclear land

5.2 Training samples

600 sample locations were used (Built-up: 200, Other: 200, Vegetation: 200), each extracted in every year, giving 1800 training rows. The PDF minimum is 50 per class. Candidates were taken only where two independent global land-cover products (ESA WorldCover 2021 and Google Dynamic World for each season) agree on the class in *every* year, so the land is stable and unambiguous. Spectral purity filters were then applied (vegetation NDVI ≥ 0.35 in all years; built-up NDVI ≤ 0.3). Samples were spread round-robin over a 5×5 block grid at least 100 m apart, so they cover the whole AOI and are not all from one small location. Built-up samples include buildings, roads and housing zones; vegetation samples include fields, trees and green patches; 'Other' covers bare land, water and unclear areas. A chip sheet (training_samples_review.pdf) is generated for checking the samples visually; any rejected sample is switched off with keep = False and the model is retrained. Validation points are at least 100 m from every training sample.

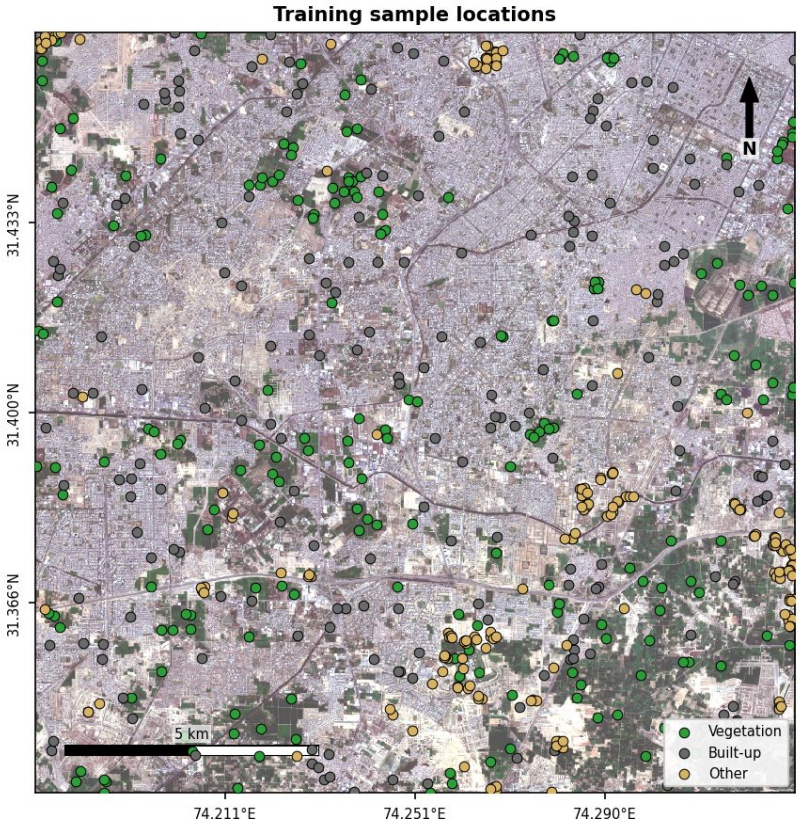


Figure - training sample locations.

5.3 Classification models

Option A (rule-based): NDVI/NDBI/MNDWI thresholds, with starting values tuned by grid search on the training samples (tuned: NDVI ≥ 0.35 → Vegetation; NDBI ≥ 0.1 → Built-up; MNDWI ≥ -0.1 → Other/water). **Option B (machine learning):** Random Forest (400 trees) and XGBoost (400 boosted trees) on 9 features: Blue, Green, Red, NIR, SWIR1, SWIR2, NDVI, NDBI, MNDWI. One model is trained on the pooled samples from all years, so the class definitions are identical in each year. Models were compared with 5-fold **spatial block cross-validation**: whole blocks are held out, which avoids the optimistic accuracy of random splits on neighbouring pixels. The best model by macro-F1 was used, followed by a 3×3 majority filter to remove salt-and-pepper noise.

Model	Overall accuracy	Macro F1	Kappa	Evaluation
Rule Based	0.674	0.644	0.511	thresholds tuned on the same samples (optimistic)
Random Forest	0.917	0.917	0.876	spatial block CV
Xgboost	0.916	0.916	0.874	spatial block CV

Selected model: **Random Forest**.

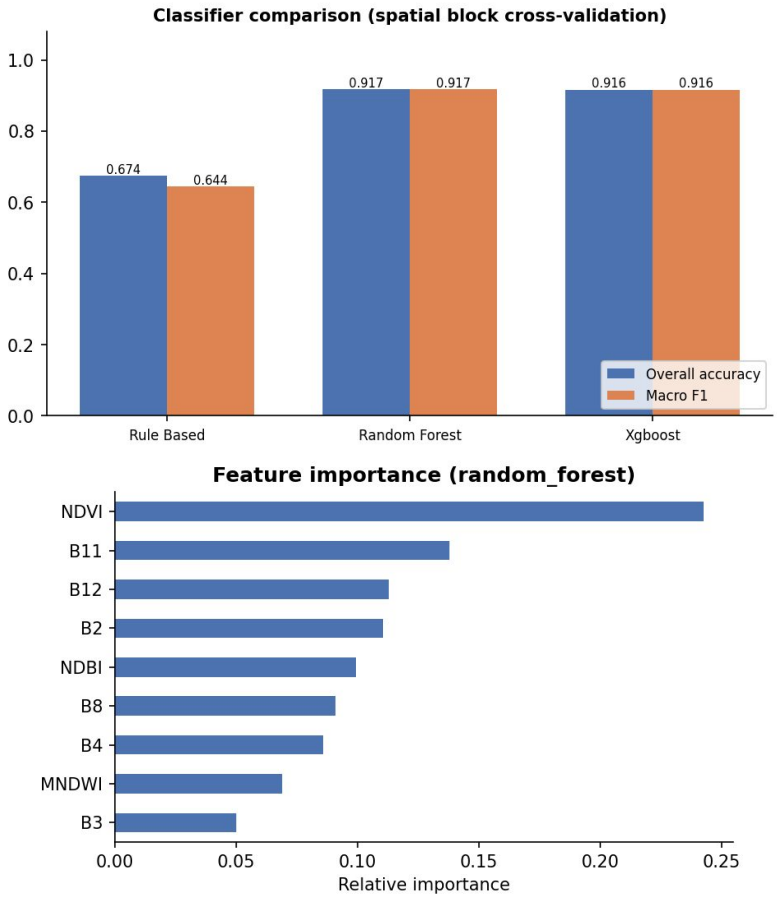


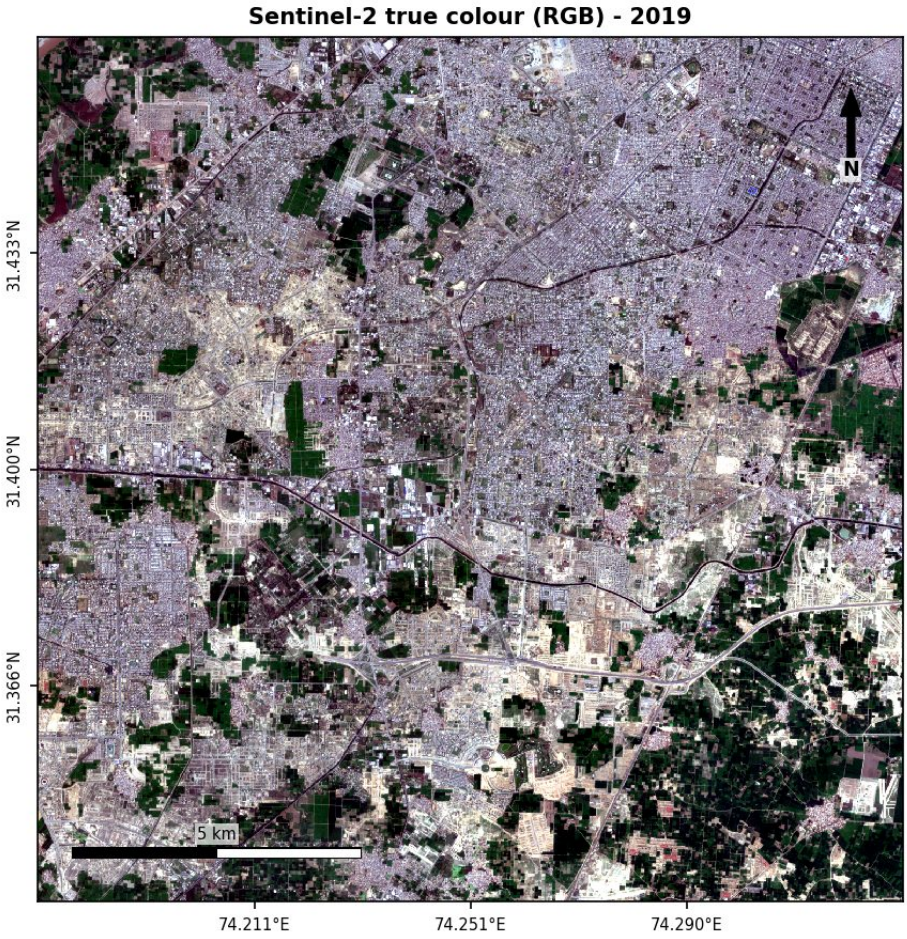
Figure - feature importance of the selected model.

5.4 Change detection and area calculation

Change was computed as described in Section 4. All areas are pixel counts × 100 m² on the 10 m UTM grid (UTM scale error inside the AOI < 0.05%). Hotspots are 1 km × 1 km cells ranked by converted area. Contiguous conversion patches are also listed. As an extra, an urban heat-risk proxy combines NDVI loss, NDBI gain and built-up share per 500 m cell. It is a relative indicator only, not a temperature measurement.

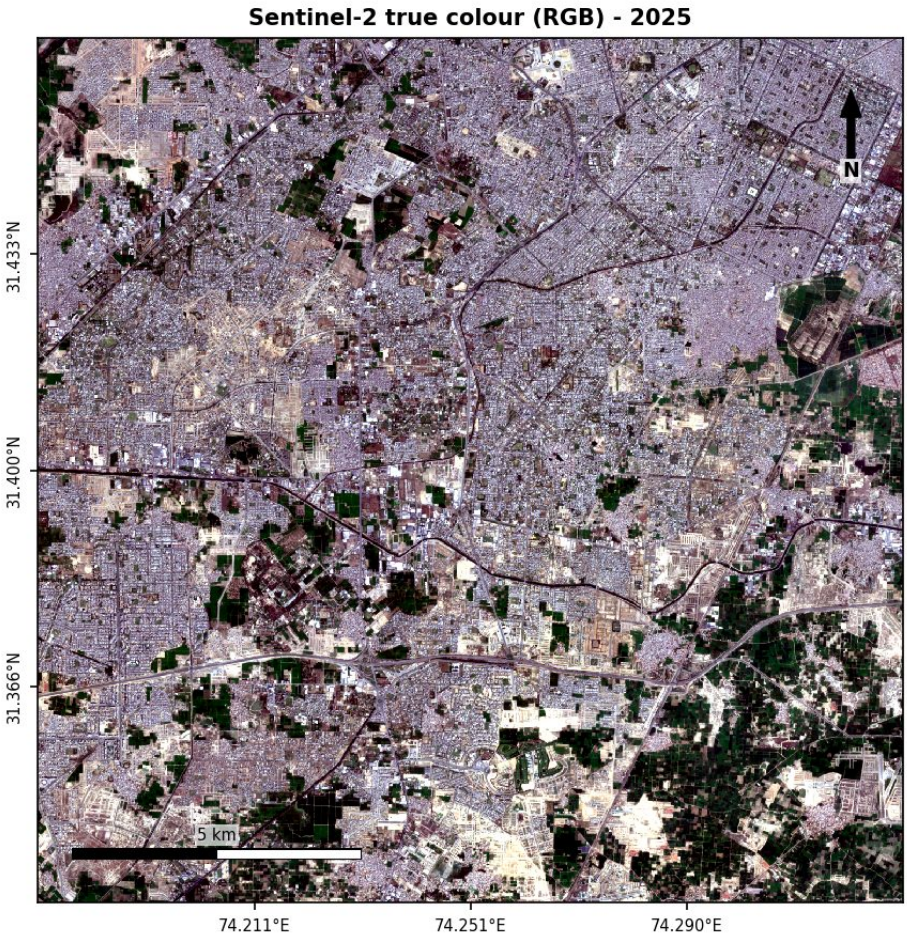
6. Results

6.1 Pre-processing outputs



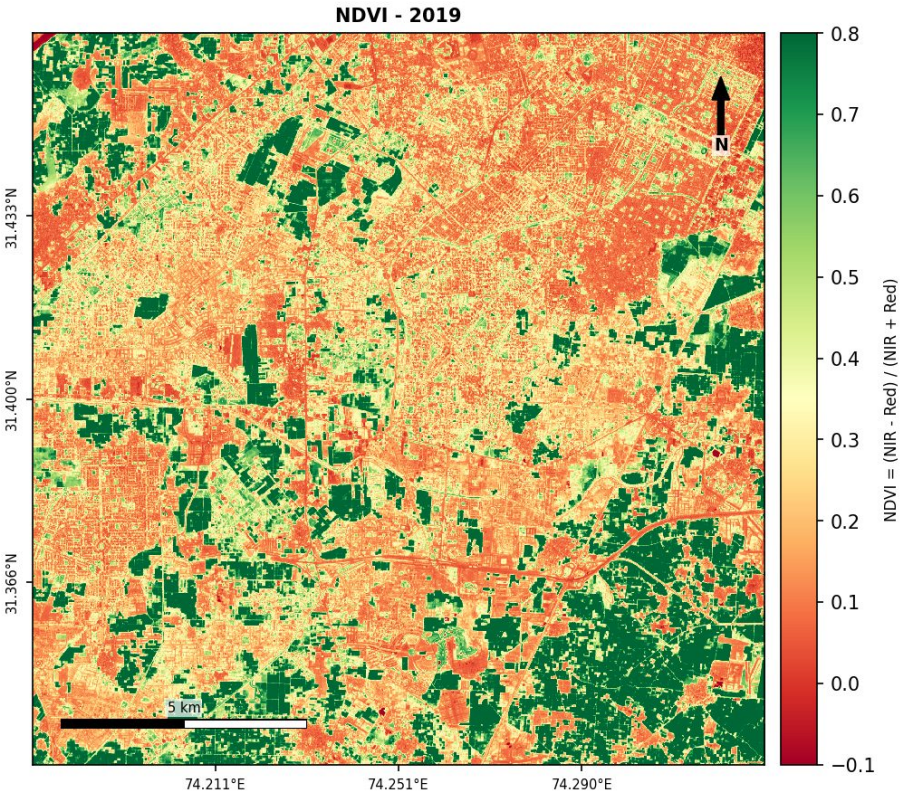
Source: Copernicus Sentinel-2 L2A (ESA), processed in Google Earth Engine. Author: Hamza Nadeem. Estimated change - learning prototype.

Figure 2 - RGB map - 2019.



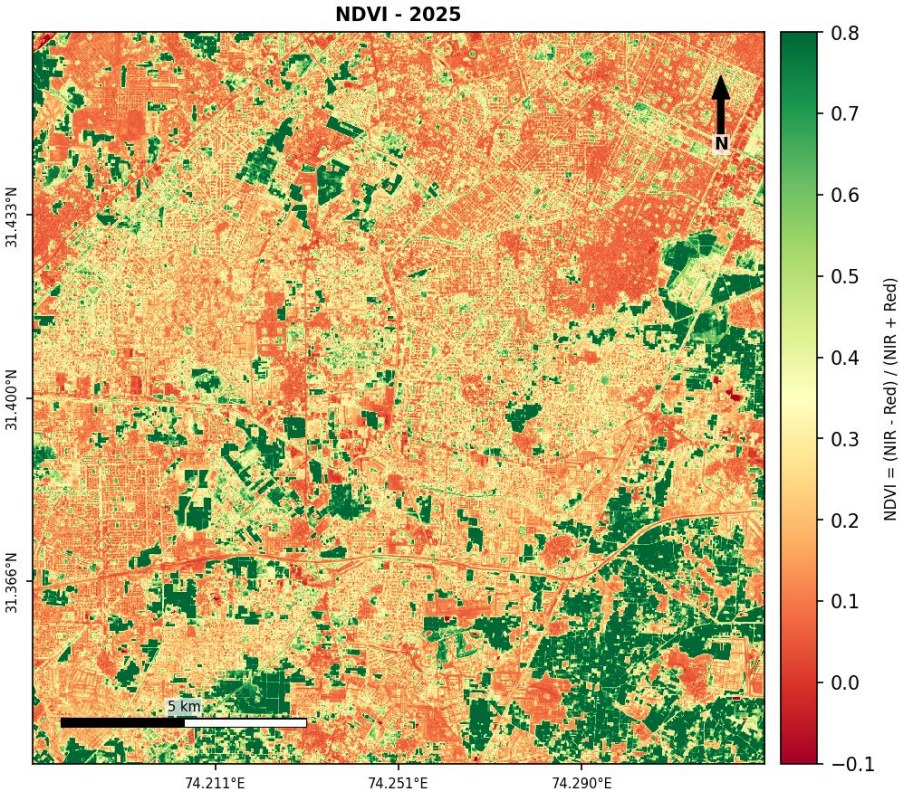
Source: Copernicus Sentinel-2 L2A (ESA), processed in Google Earth Engine. Author: Hamza Nadeem. Estimated change - learning prototype.

Figure 3 - RGB map - 2025.



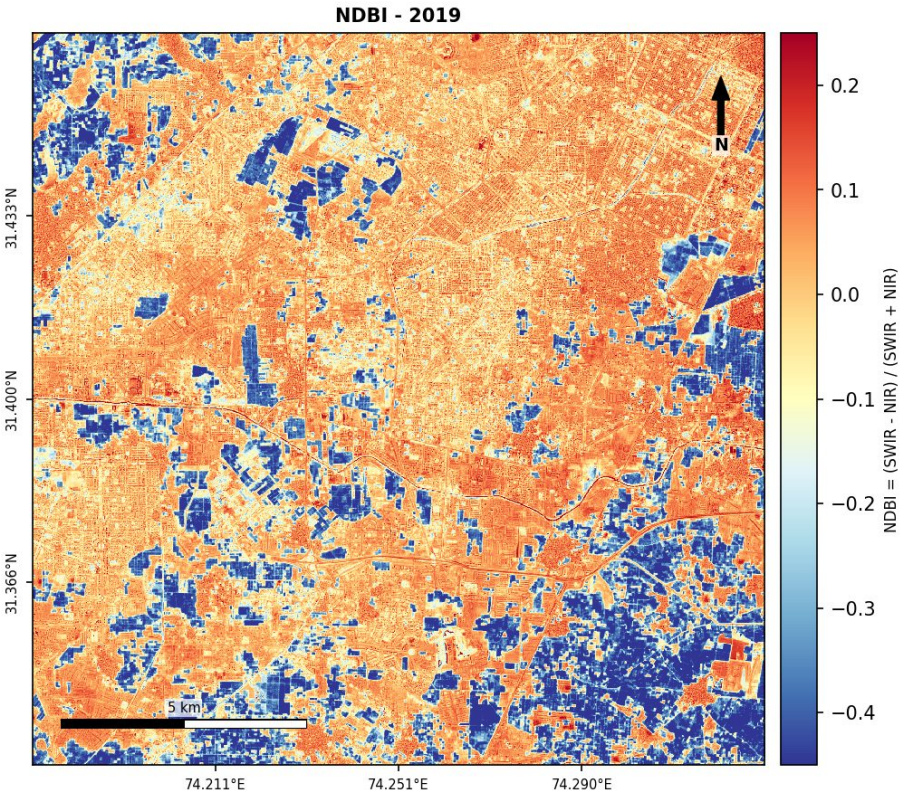
Source: Copernicus Sentinel-2 L2A (ESA), processed in Google Earth Engine. Author: Hamza Nadeem. Estimated change - learning prototype.

Figure 4 - NDVI map - 2019.



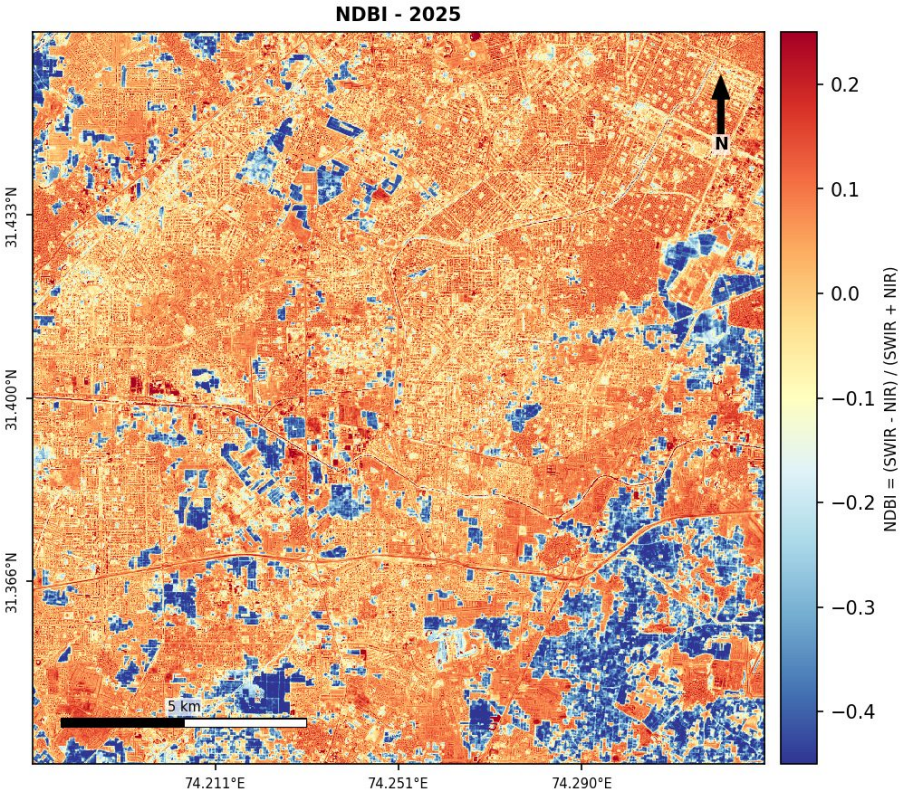
Source: Copernicus Sentinel-2 L2A (ESA), processed in Google Earth Engine. Author: Hamza Nadeem. Estimated change - learning prototype.

Figure 5 - NDVI map - 2025.



Source: Copernicus Sentinel-2 L2A (ESA), processed in Google Earth Engine. Author: Hamza Nadeem. Estimated change - learning prototype.

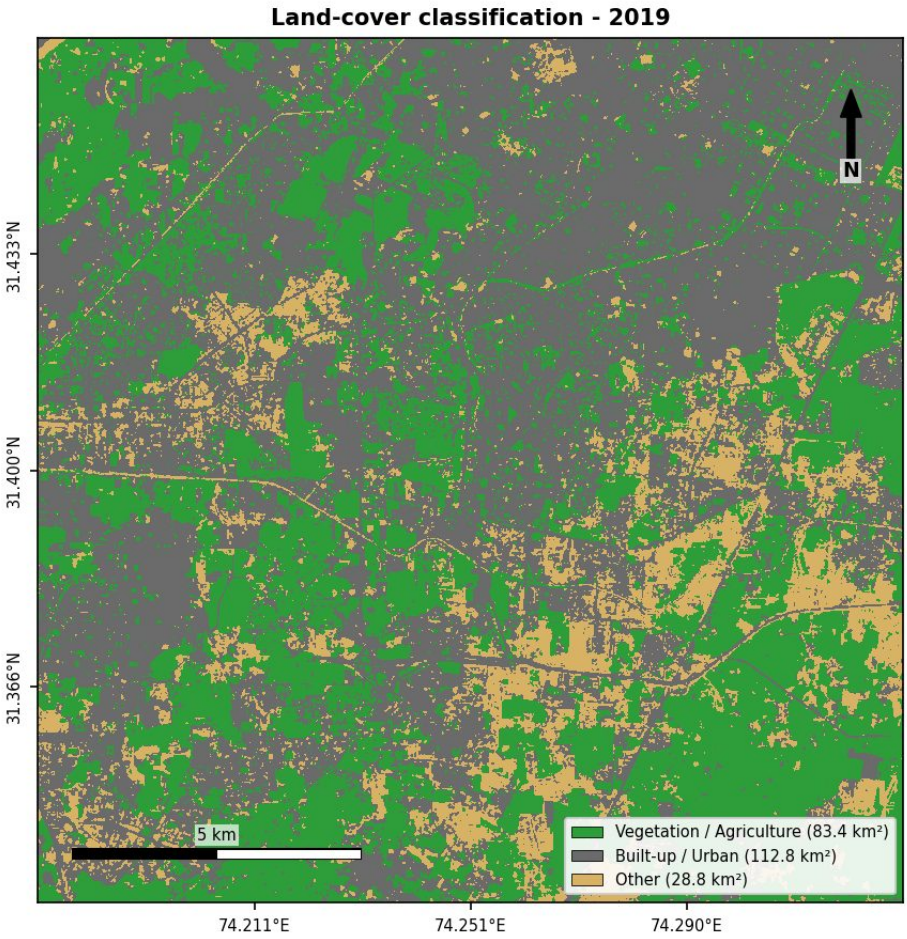
Figure 6 - NDBI map - 2019.



Source: Copernicus Sentinel-2 L2A (ESA), processed in Google Earth Engine. Author: Hamza Nadeem. Estimated change - learning prototype.

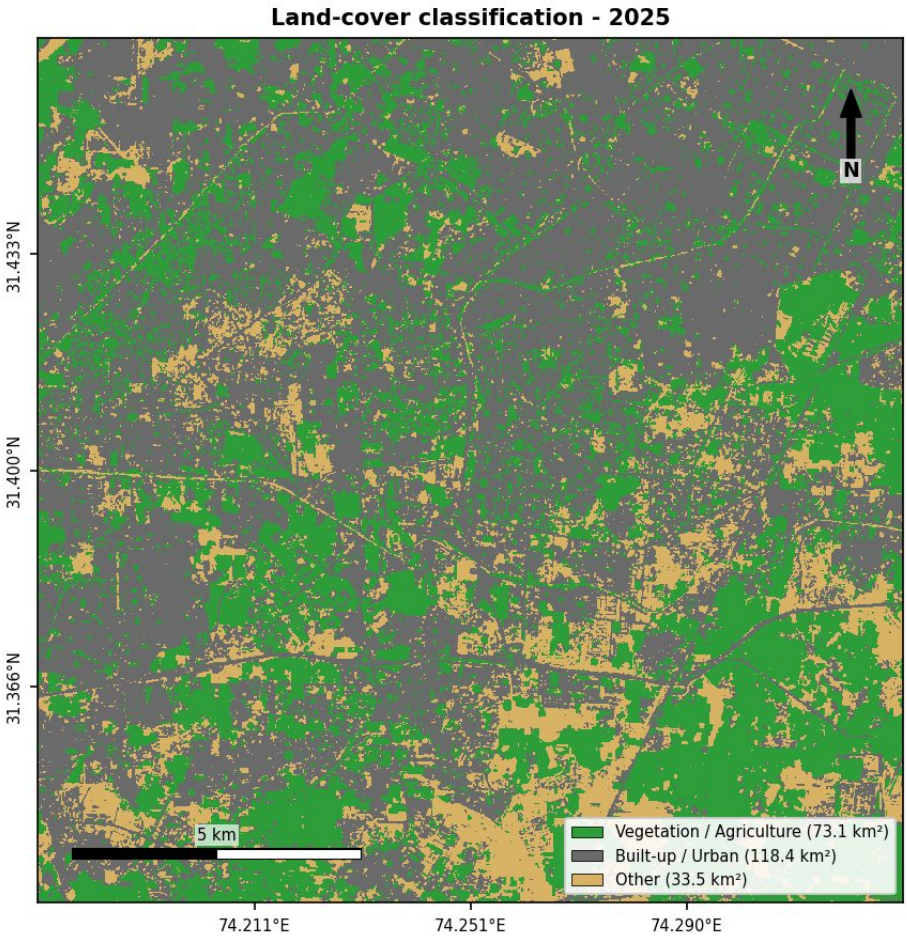
Figure 7 - NDBI map - 2025.

6.2 Land-cover classification



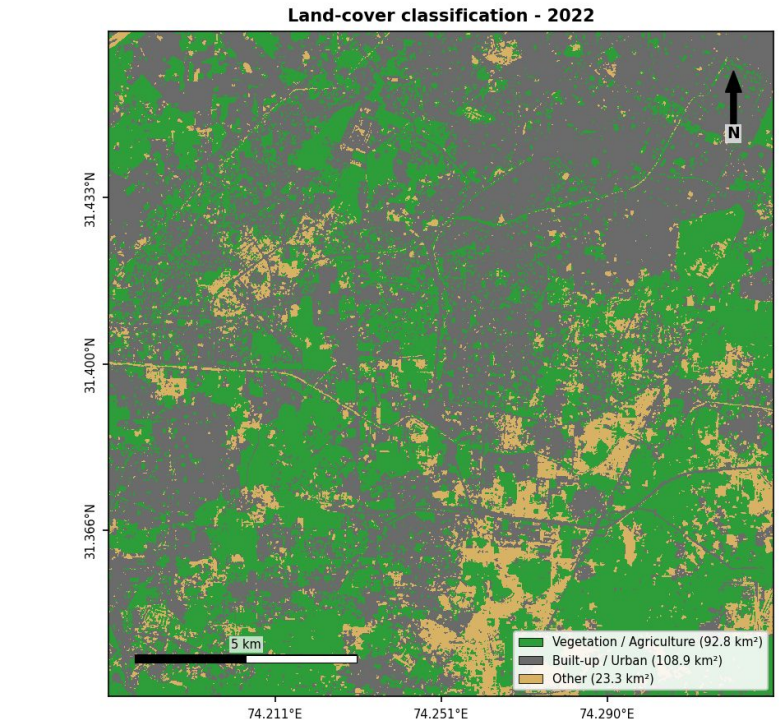
Source: Copernicus Sentinel-2 L2A (ESA), processed in Google Earth Engine. Author: Hamza Nadeem. Estimated change - learning prototype.

Figure 8 - Classified land-cover map - 2019.



Source: Copernicus Sentinel-2 L2A (ESA), processed in Google Earth Engine. Author: Hamza Nadeem. Estimated change - learning prototype.

Figure 9 - Classified land-cover map - 2025.



Source: Copernicus Sentinel-2 L2A (ESA), processed in Google Earth Engine. Author: Hamza Nadeem. Estimated change - learning prototype.

Figure 10 - Land cover 2022 (extra year for the trend).

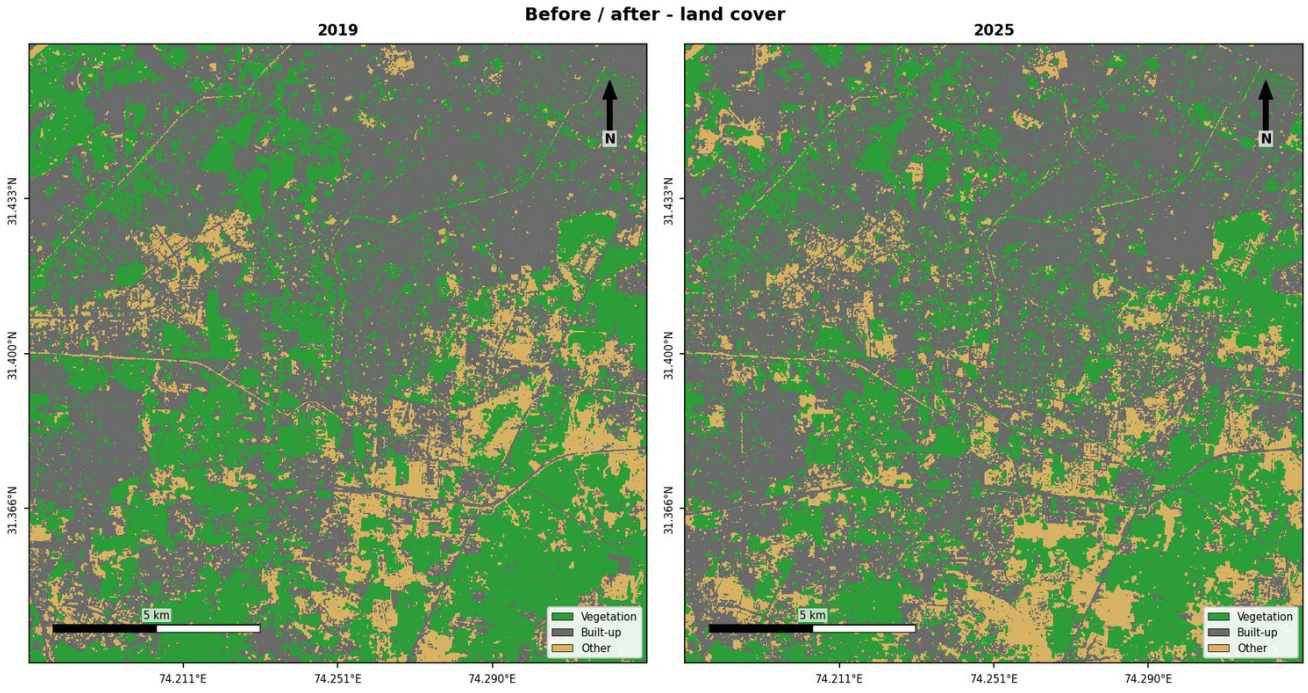
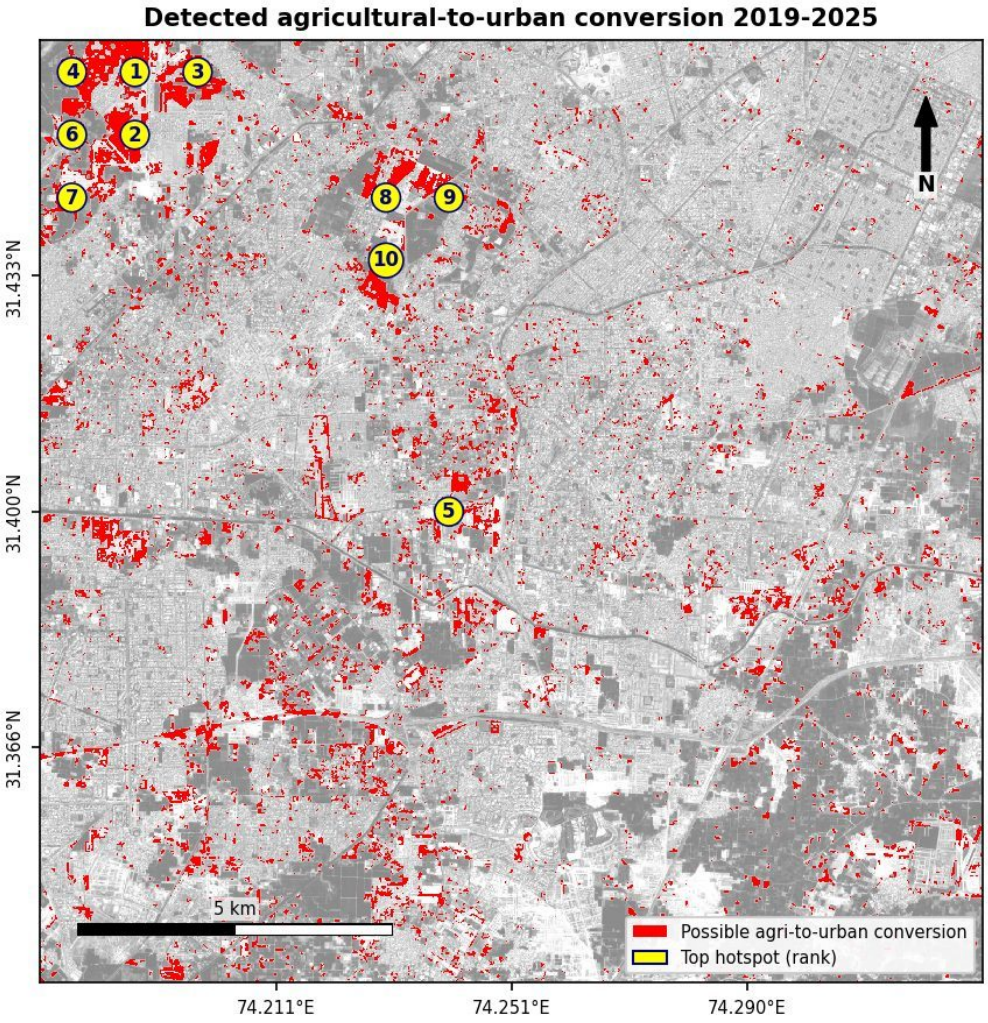


Figure 11 - Land cover before / after.

6.3 Change detection



Source: Copernicus Sentinel-2 L2A (ESA), processed in Google Earth Engine. Author: Hamza Nadeem. Estimated change - learning prototype.

Figure 12 - Change-detection map: possible agricultural-to-urban conversion 2019-2025 (bright red), top hotspots numbered.

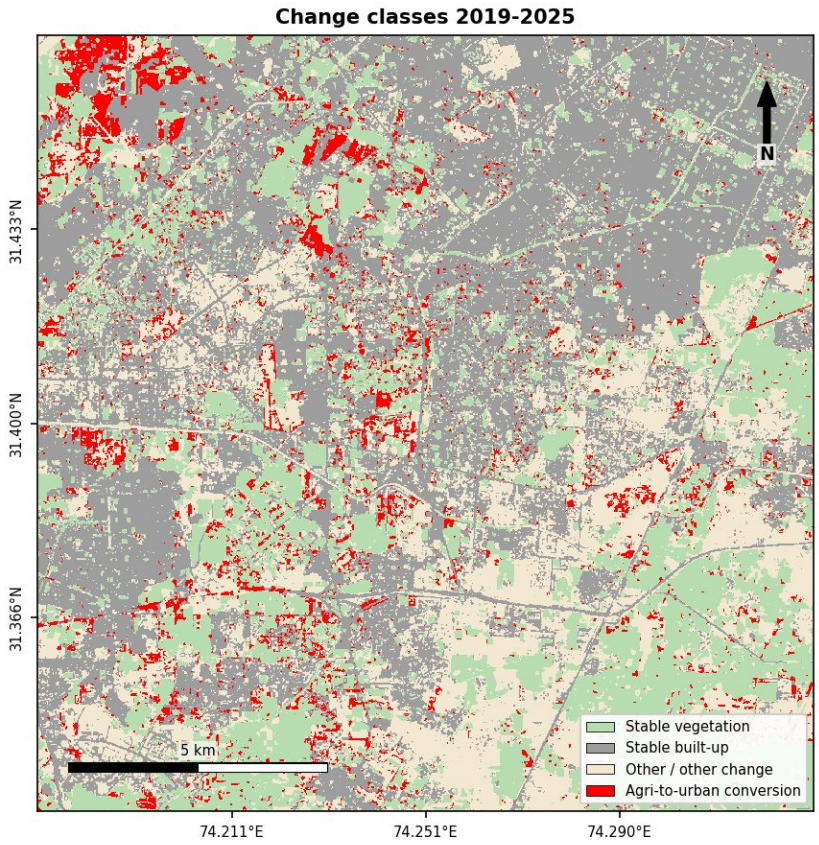


Figure 13 - Change classes.

6.4 Area statistics

Metric	Value
Total AOI Area	225.00 sq. km
Area analysed (cloud-free both years)	225.00 sq. km
Vegetation / Agriculture Area in 2019	83.40 sq. km
Vegetation / Agriculture Area in 2025	73.09 sq. km
Built-up Area in 2019	112.82 sq. km
Built-up Area in 2025	118.42 sq. km
Net Built-up Growth	+5.60 sq. km
Agriculture-to-Urban Conversion (estimated)	12.31 sq. km (1231 ha)
Percentage of AOI Converted	5.47 %
Share of 2019 Vegetation Converted	14.76 %
Low-confidence part of conversion	2.92 sq. km

Transition matrix 2019 → 2025 (km²):

From 2019 \ To 2025	Vegetation 2025	Built-up 2025	Other 2025
Vegetation 2019	57.495	14.402	11.499
Built-up 2019	11.35	93.017	8.452
Other 2019	4.245	11.005	13.536

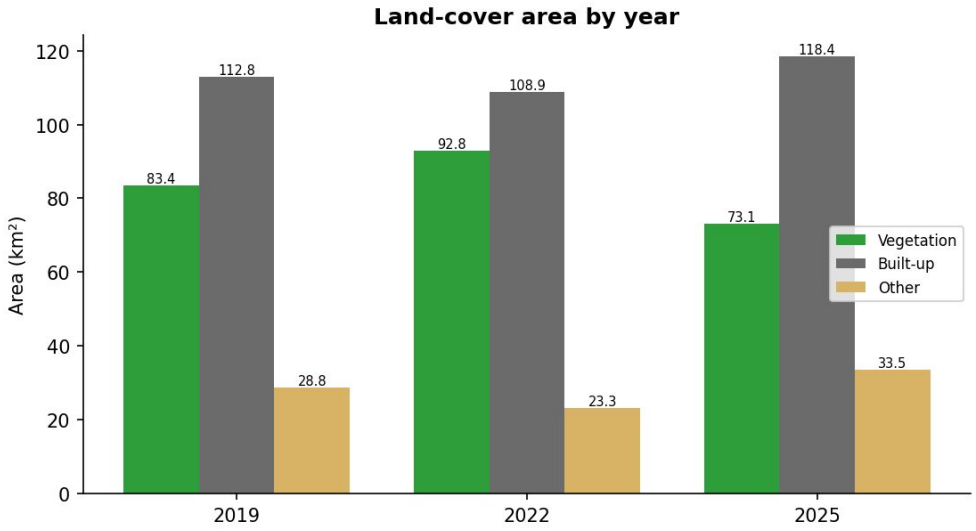


Figure 14 - Area by class and year.

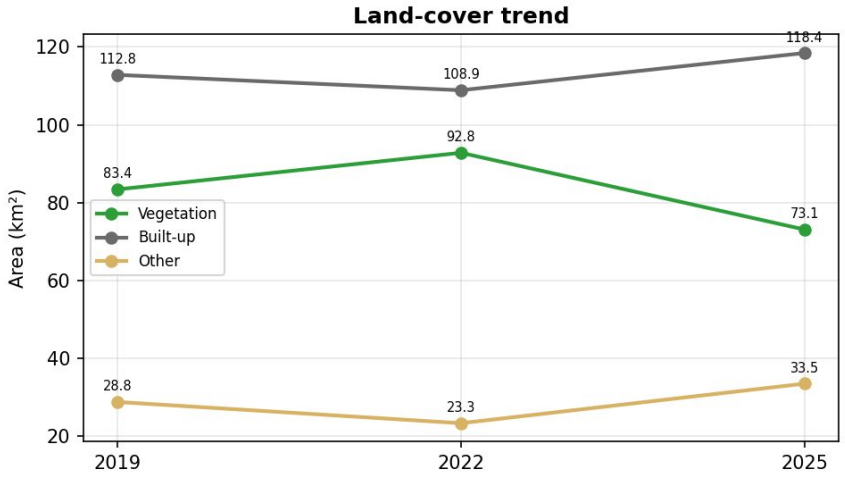


Figure 15 - Land-cover trend.

6.5 Change hotspots

- Hotspot #1: about 46.1 ha of possible conversion (46.0% of the 1 km² cell) in the north-western part of the AOI (9.2 km from centre) [31.4627°N, 74.1864°E].
- Hotspot #2: about 35.8 ha of possible conversion (35.8% of the 1 km² cell) in the north-western part of the AOI (8.5 km from centre) [31.4537°N, 74.1865°E].
- Hotspot #3: about 34.3 ha of possible conversion (34.3% of the 1 km² cell) in the north-western part of the AOI (8.6 km from centre) [31.4628°N, 74.1969°E].
- Hotspot #4: about 30.8 ha of possible conversion (30.8% of the 1 km² cell) in the north-western part of the AOI (9.9 km from centre) [31.4627°N, 74.1759°E].
- Hotspot #5: about 21.9 ha of possible conversion (21.9% of the 1 km² cell) in the central part of the AOI (1.0 km from centre) [31.3999°N, 74.2395°E].
- The top 5 one-square-kilometre cells hold 14% of all detected conversion, so the change is spread across many locations.

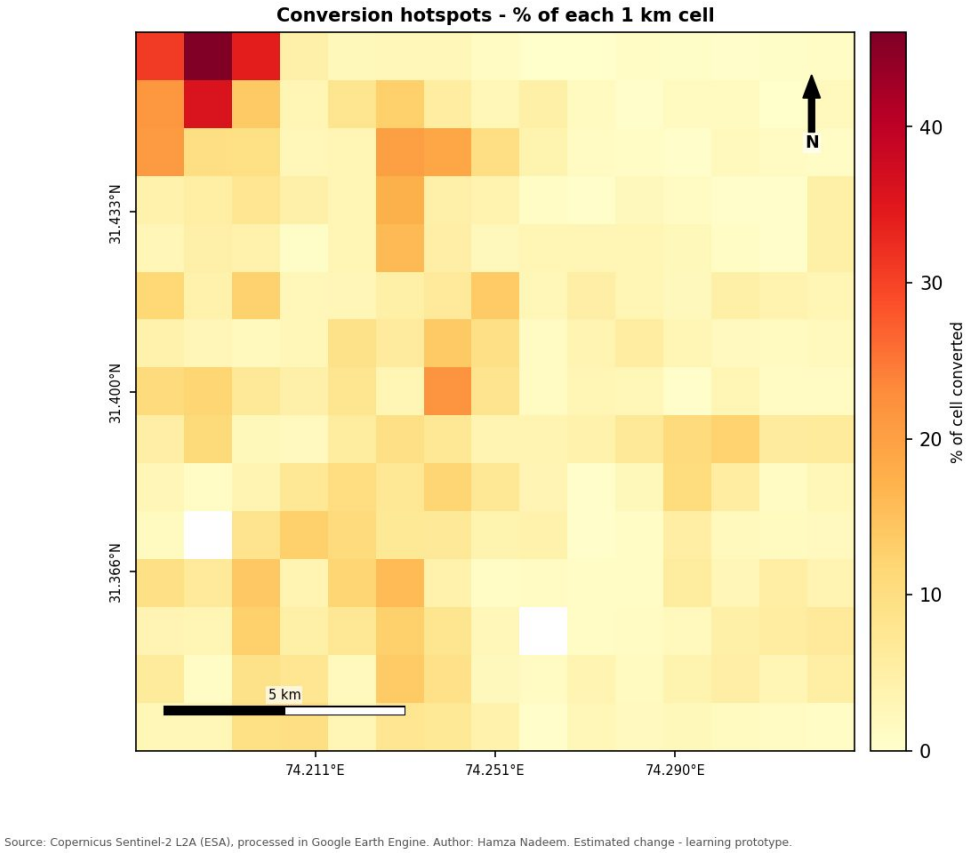
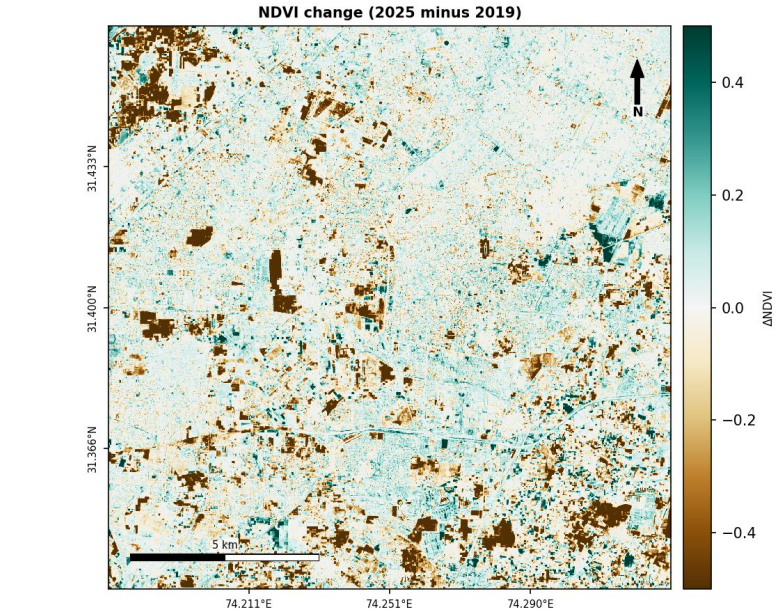


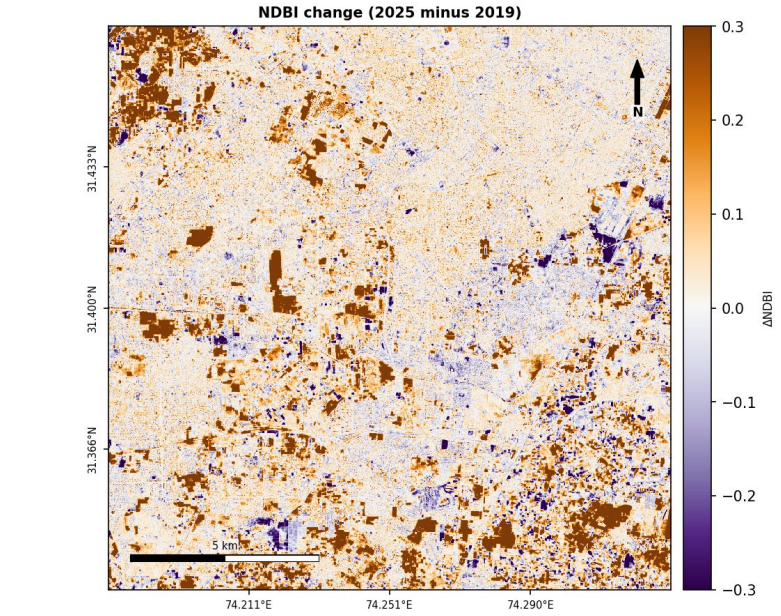
Figure 16 - Hotspot density.

Rank	Converted (ha)	% of 1 km² cell	Lat	Lon
1	46.0	46.0	31.46275	74.18640
2	35.8	35.8	31.45373	74.18648
3	34.3	34.3	31.46282	74.19692
4	30.8	30.8	31.46269	74.17587
5	21.9	21.9	31.39992	74.23953
6	21.2	21.2	31.45367	74.17595
7	20.8	20.8	31.44464	74.17603
8	20.1	20.1	31.44497	74.22865
9	18.9	18.9	31.44503	74.23917
10	17.3	17.3	31.43595	74.22872



Source: Copernicus Sentinel-2 L2A (ESA), processed in Google Earth Engine. Author: Hamza Nadeem. Estimated change - learning prototype.

Figure 17 - NDVI change.



Source: Copernicus Sentinel-2 L2A (ESA), processed in Google Earth Engine. Author: Hamza Nadeem. Estimated change - learning prototype.

Figure 18 - NDBI change.

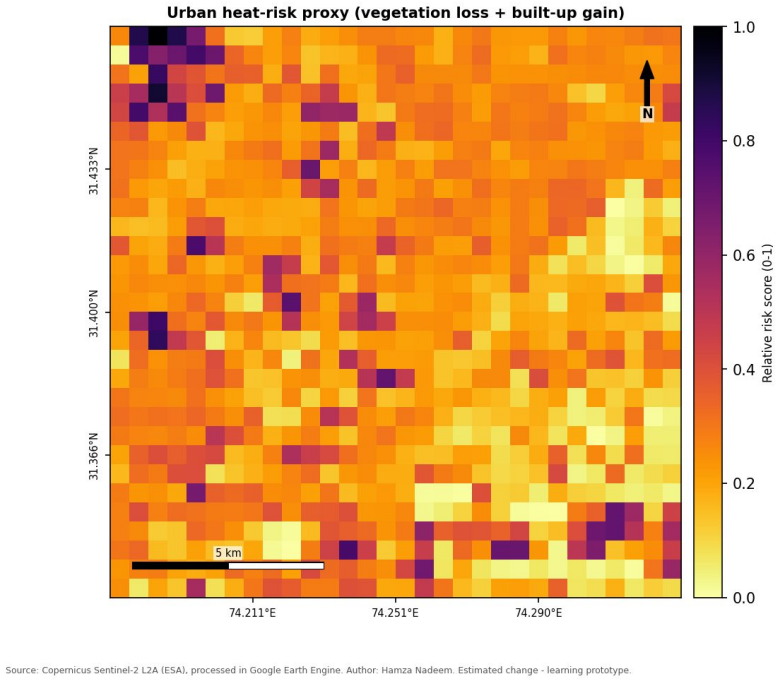


Figure 19 - Urban heat-risk proxy (relative, not temperature).

7. Validation

60 validation points (PDF minimum: 50) were drawn by stratified random sampling from the 2025 map: 15 per stratum (vegetation, built-up, other, detected conversion). Points are at least 200 m apart and at least 100 m from any training sample. Each point was checked visually without seeing the model prediction, using 2019/2025 Sentinel-2 chips (true and false colour) and the Google Maps high-resolution satellite basemap. The class visible at the point was recorded, along with whether vegetation had become built-up.

Metric	Value
Points checked	60
Correct predictions	29
Incorrect predictions	31
Approximate validation accuracy	48.3 %
Area-weighted overall accuracy	56.6 %
Cohen's kappa	0.114
Conversion user's accuracy (precision)	0.467
Conversion producer's accuracy (recall)	0.500
Error-adjusted conversion area (95% CI)	33.70 km² (10.92-56.47)
Rule-based (Option A) accuracy on the same points	20.0 %

The area-weighted accuracy corrects for the equal number of points per stratum. The error-adjusted area uses the stratified estimator of Olofsson et al. (2014), the standard good-practice method for estimating area from a map plus reference sample.

Class	Reference pts	Mapped pts	Producer's acc.	User's acc.
Vegetation	12	15	0.5	0.4
Built-up	40	30	0.5	0.667
Other	8	15	0.375	0.2

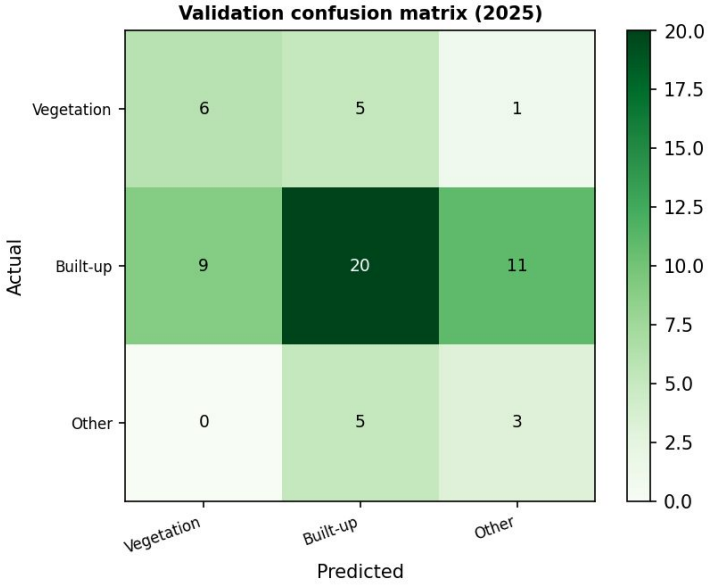


Figure - validation confusion matrix.

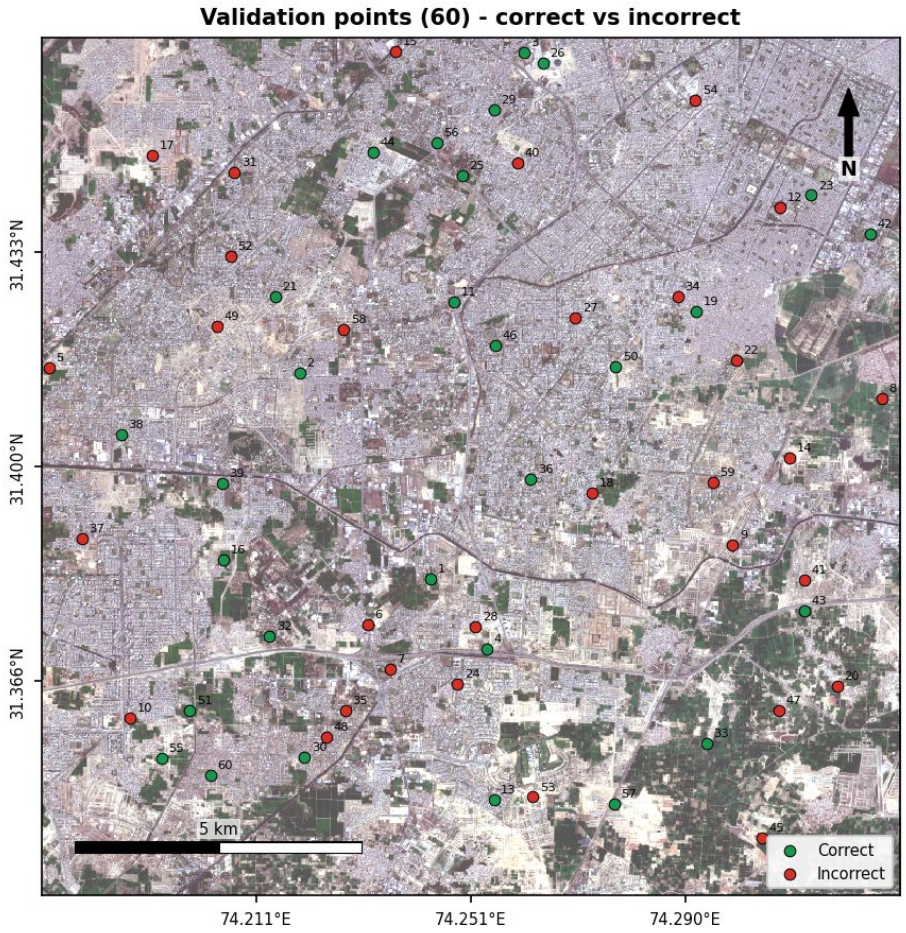


Figure - validation points (green = correct, red = incorrect).

Validation table

Point ID	Actual Class	Predicted Class	Correct / Incorrect	Actual Change	Predicted Change	Notes
1	Built-up	Built-up	Correct	yes	yes	new houses on former farmland new housing scheme, was crop field in 2019 new road built on green land houses with some trees, was field in 2019 new building / construction completed
2	Built-up	Built-up	Correct	no	yes	old built-up area, no change existing houses since 2019 existing road dense town area, no change
3	Built-up	Built-up	Correct	no	no	old built-up area, no change existing houses since 2019 existing road dense town area, no change
4	Other	Other	Correct	no	no	bare/fallow field, no buildings cleared plot, construction not started harvested field, bare soil canal / water brick kiln / open dusty area unclear image
5	Vegetation	Built-up	Incorrect	no	yes	crop field, no change wheat field / green crops trees / orchard green but less dense than 2019 park / green belt
6	Built-up	Other	Incorrect	yes	no	
7	Other	Built-up	Incorrect	no	no	
8	Vegetation	Other	Incorrect	no	no	
9	Other	Built-up	Incorrect	no	yes	
10	Built-up	Other	Incorrect	no	no	
11	Other	Other	Correct	no	no	
12	Built-up	Other	Incorrect	no	no	
13	Built-up	Built-up	Correct	no	no	
14	Other	Built-up	Incorrect	yes	yes	

Point ID	Actual Class	Predicted Class	Correct / Incorrect	Actual Change	Predicted Change	Notes
15	Built-up	Other	Incorrect	no	no	
16	Vegetation	Vegetation	Correct	no	no	
17	Other	Built-up	Incorrect	yes	yes	
18	Vegetation	Built-up	Incorrect	no	yes	
19	Built-up	Built-up	Correct	no	no	
20	Vegetation	Built-up	Incorrect	no	yes	
21	Built-up	Built-up	Correct	no	no	
22	Built-up	Vegetation	Incorrect	no	no	
23	Built-up	Built-up	Correct	no	no	
24	Built-up	Other	Incorrect	no	no	
25	Vegetation	Vegetation	Correct	no	no	
26	Other	Other	Correct	no	no	
27	Built-up	Vegetation	Incorrect	no	no	
28	Built-up	Vegetation	Incorrect	yes	no	
29	Built-up	Built-up	Correct	no	yes	
30	Built-up	Built-up	Correct	no	no	
31	Built-up	Vegetation	Incorrect	no	no	
32	Vegetation	Vegetation	Correct	no	no	
33	Vegetation	Vegetation	Correct	no	no	
34	Built-up	Other	Incorrect	no	no	
35	Vegetation	Built-up	Incorrect	no	yes	
36	Built-up	Built-up	Correct	yes	no	
37	Built-up	Other	Incorrect	yes	no	
38	Built-up	Built-up	Correct	no	no	
39	Built-up	Built-up	Correct	yes	no	
40	Built-up	Vegetation	Incorrect	no	no	
41	Other	Built-up	Incorrect	no	no	
42	Built-up	Built-up	Correct	no	yes	
43	Vegetation	Vegetation	Correct	no	no	
44	Built-up	Built-up	Correct	yes	yes	
45	Built-up	Other	Incorrect	yes	no	
46	Built-up	Built-up	Correct	no	no	
47	Vegetation	Built-up	Incorrect	no	no	
48	Built-up	Vegetation	Incorrect	no	no	
49	Built-up	Other	Incorrect	no	no	
50	Built-up	Built-up	Correct	yes	yes	
51	Vegetation	Vegetation	Correct	no	no	
52	Built-up	Vegetation	Incorrect	no	no	
53	Built-up	Other	Incorrect	yes	no	
54	Built-up	Vegetation	Incorrect	no	no	
55	Built-up	Built-up	Correct	yes	yes	
56	Built-up	Built-up	Correct	no	no	
57	Built-up	Built-up	Correct	yes	yes	
58	Built-up	Vegetation	Incorrect	no	no	
59	Built-up	Other	Incorrect	no	no	
60	Built-up	Built-up	Correct	no	no	

8. Error Analysis

36 weak or incorrect cases were analysed (PDF minimum: 15): 36 incorrect results plus the lowest-confidence correct results. Error types and possible reasons are suggested automatically from the pixel evidence (NDVI/NDBI in both years, clear observations, model confidence). They are hypotheses to be confirmed against each point's image chip; analyst comments appear in the Notes column of the validation table.

Case	Location / Point	Expected Result	Model Result	Error Type	Possible Reason (suggested)	Evidence
1	Point 5	Vegetation (no change)	Conversion detected	False conversion	Bare / fallow soil confused with built-up	NDVI 2019=0.50, 2025=0.23; NDBI 2025=-0.02; clear obs=8; confidence=0.62
2	Point 6	Vegetation-to-built-up	Other (no change)	Missed change	New road / construction site confused with barren land	NDVI 2019=0.52, 2025=0.06; NDBI 2025=0.00; clear obs=10; confidence=0.63
3	Point 37	Vegetation-to-built-up	Other (no change)	Missed change	New road / construction site confused with barren land	NDVI 2019=0.57, 2025=0.29; NDBI 2025=0.05; clear obs=9; confidence=0.63
4	Point 9	Other (no change)	Conversion detected	False conversion	Seasonal crop change	NDVI 2019=0.39, 2025=0.25; NDBI 2025=0.09; clear obs=5; confidence=0.75
5	Point 45	Vegetation-to-built-up	Other (no change)	Missed change	New road / construction site confused with barren land	NDVI 2019=0.85, 2025=0.18; NDBI 2025=0.01; clear obs=4; confidence=0.86
6	Point 35	Vegetation (no change)	Conversion detected	False conversion	Bare / fallow soil confused with built-up	NDVI 2019=0.48, 2025=0.23; NDBI 2025=0.11; clear obs=10; confidence=0.91
7	Point 20	Vegetation (no change)	Conversion detected	False conversion	Bare / fallow soil confused with built-up	NDVI 2019=0.85, 2025=0.23; NDBI 2025=0.13; clear obs=5; confidence=0.94
8	Point 28	Vegetation-to-built-up	Vegetation (no change)	Missed change	Tree-covered housing area classified as vegetation	NDVI 2019=0.50, 2025=0.38; NDBI 2025=-0.01; clear obs=9; confidence=0.96
9	Point 18	Vegetation (no change)	Conversion detected	False conversion	Bare / fallow soil confused with built-up	NDVI 2019=0.69, 2025=0.12; NDBI 2025=0.07; clear obs=5; confidence=0.96
10	Point 53	Vegetation-to-built-up	Other (no change)	Missed change	New road / construction site confused with barren land	NDVI 2019=0.92, 2025=0.21; NDBI 2025=0.05; clear obs=4; confidence=1.00
11	Point 29	Built-up (no change)	Conversion detected	False conversion	Change already present in baseline map	NDVI 2019=0.56, 2025=0.31; NDBI 2025=0.03; clear obs=10; confidence=0.41
12	Point 39	Vegetation-to-built-up	Built-up (no change)	Missed change	Small buildings / mixed pixels at 10 m	NDVI 2019=0.09, 2025=0.15; NDBI 2025=0.05; clear obs=9; confidence=0.67
13	Point 36	Vegetation-to-built-up	Built-up (no change)	Missed change	Baseline-year map error (old year mapped as built-up)	NDVI 2019=0.20, 2025=0.10; NDBI 2025=-0.11; clear obs=7; confidence=0.73
14	Point 2	Built-up (no change)	Conversion detected	False conversion	Change already present in baseline map	NDVI 2019=0.53, 2025=0.35; NDBI 2025=0.03; clear obs=8; confidence=0.83
15	Point 42	Built-up (no change)	Conversion detected	False conversion	Change already present in baseline map	NDVI 2019=0.37, 2025=0.19; NDBI 2025=0.06; clear obs=4; confidence=0.91

Case	Location / Point	Expected Result	Model Result	Error Type	Possible Reason (suggested)	Evidence
16	Point 15	Built-up (no change)	Other (no change)	Missed urban area	New road / construction site confused with barren land	NDVI 2019=0.35, 2025=0.47; NDBI 2025=-0.01; clear obs=10; confidence=0.38
17	Point 48	Built-up (no change)	Vegetation (no change)	Missed urban area	Tree-covered housing area classified as vegetation	NDVI 2019=0.31, 2025=0.34; NDBI 2025=0.03; clear obs=9; confidence=0.40
18	Point 34	Built-up (no change)	Other (no change)	Missed urban area	New road / construction site confused with barren land	NDVI 2019=0.28, 2025=0.35; NDBI 2025=0.01; clear obs=4; confidence=0.43
19	Point 22	Built-up (no change)	Vegetation (no change)	Missed urban area	Tree-covered housing area classified as vegetation	NDVI 2019=0.31, 2025=0.64; NDBI 2025=-0.21; clear obs=4; confidence=0.44
20	Point 41	Other (no change)	Built-up (no change)	False built-up	Bare / fallow soil confused with built-up	NDVI 2019=0.15, 2025=0.30; NDBI 2025=0.10; clear obs=5; confidence=0.47
21	Point 10	Built-up (no change)	Other (no change)	Missed urban area	New road / construction site confused with barren land	NDVI 2019=0.31, 2025=0.15; NDBI 2025=0.00; clear obs=10; confidence=0.53
22	Point 49	Built-up (no change)	Other (no change)	Missed urban area	New road / construction site confused with barren land	NDVI 2019=0.12, 2025=0.10; NDBI 2025=0.12; clear obs=9; confidence=0.53
23	Point 59	Built-up (no change)	Other (no change)	Missed urban area	New road / construction site confused with barren land	NDVI 2019=0.23, 2025=0.31; NDBI 2025=0.05; clear obs=5; confidence=0.57
24	Point 14	Vegetation-to-built-up	Conversion detected	False built-up	Bare / fallow soil confused with built-up	NDVI 2019=0.47, 2025=0.15; NDBI 2025=0.12; clear obs=5; confidence=0.57
25	Point 24	Built-up (no change)	Other (no change)	Missed urban area	New road / construction site confused with barren land	NDVI 2019=0.14, 2025=0.17; NDBI 2025=0.02; clear obs=9; confidence=0.59
26	Point 40	Built-up (no change)	Vegetation (no change)	Missed urban area	Tree-covered housing area classified as vegetation	NDVI 2019=0.32, 2025=0.43; NDBI 2025=-0.13; clear obs=10; confidence=0.60
27	Point 47	Vegetation (no change)	Built-up (no change)	False built-up	Bare / fallow soil confused with built-up	NDVI 2019=0.09, 2025=0.09; NDBI 2025=0.13; clear obs=5; confidence=0.66
28	Point 17	Vegetation-to-built-up	Conversion detected	False built-up	Bare / fallow soil confused with built-up	NDVI 2019=0.54, 2025=0.08; NDBI 2025=0.09; clear obs=10; confidence=0.73
29	Point 12	Built-up (no change)	Other (no change)	Missed urban area	New road / construction site confused with barren land	NDVI 2019=0.24, 2025=0.20; NDBI 2025=0.01; clear obs=4; confidence=0.76
30	Point 7	Other (no change)	Built-up (no change)	False built-up	Bare / fallow soil confused with built-up	NDVI 2019=0.09, 2025=0.08; NDBI 2025=0.11; clear obs=9; confidence=0.77
31	Point 54	Built-up (no change)	Vegetation (no change)	Missed urban area	Tree-covered housing area classified as vegetation	NDVI 2019=0.50, 2025=0.37; NDBI 2025=0.10; clear obs=4; confidence=0.77
32	Point 52	Built-up (no change)	Vegetation (no change)	Missed urban area	Tree-covered housing area classified as vegetation	NDVI 2019=0.40, 2025=0.44; NDBI 2025=0.02; clear obs=9; confidence=0.88
33	Point 8	Vegetation (no change)	Other (no change)	Wrong class	Seasonal crop change	NDVI 2019=0.87, 2025=0.21; NDBI 2025=0.00; clear obs=4; confidence=0.90

Case	Location / Point	Expected Result	Model Result	Error Type	Possible Reason (suggested)	Evidence
34	Point 31	Built-up (no change)	Vegetation (no change)	Missed urban area	Tree-covered housing area classified as vegetation	NDVI 2019=0.36, 2025=0.38; NDBI 2025=-0.11; clear obs=10; confidence=0.91
35	Point 27	Built-up (no change)	Vegetation (no change)	Missed urban area	Tree-covered housing area classified as vegetation	NDVI 2019=0.39, 2025=0.43; NDBI 2025=0.01; clear obs=7; confidence=0.97
36	Point 58	Built-up (no change)	Vegetation (no change)	Missed urban area	Tree-covered housing area classified as vegetation	NDVI 2019=0.62, 2025=0.61; NDBI 2025=-0.09; clear obs=8; confidence=1.00

Most common error sources

- New road / construction site confused with barren land: 11 case(s)
- Bare / fallow soil confused with built-up: 9 case(s)
- Tree-covered housing area classified as vegetation: 9 case(s)
- Change already present in baseline map: 3 case(s)
- Seasonal crop change: 2 case(s)
- Small buildings / mixed pixels at 10 m: 1 case(s)
- Baseline-year map error (old year mapped as built-up): 1 case(s)

Known limitations that affect all results: cloud cover and haze, seasonal crop changes (harvested or fallow fields), bare soil confused with built-up, new roads confused with barren land, small buildings missed at 10 m, tree-covered housing classified as vegetation, different image dates within the season, possibly imperfect training samples or thresholds, and mixed pixels.

Suggested improvements

- Add a separate 'Construction / bare urban' class and a road-network layer (OSM).
- Add more 'Other' training samples on harvested/fallow fields and use a multi-date (e.g. Feb + Apr) composite so fields that green up at any time are recognised.
- Add texture features (e.g. GLCM on NIR) or higher-resolution imagery to detect rooftops under tree canopy.
- Improve the baseline-year map with samples drawn from baseline imagery only, and validate the baseline year separately.
- Compare images from the same crop stage, or use several dates per year (seasonal NDVI maximum) instead of a single seasonal median.
- Use higher-resolution imagery (PlanetScope / aerial) or a U-Net segmentation model trained on building footprints.
- Check the baseline year separately: add baseline-only training samples in green peri-urban plots and compare against historical high-resolution imagery.

9. Answers to the Key Questions

1. How much land has changed from vegetation/agriculture to built-up area?

An estimated 12.31 km² (1231 ha), 5.47% of the AOI, between 2019 and 2025. With validation errors taken into account, the estimate is 33.70 ± 22.78 km².

2. Where are the major change hotspots?

Hotspot #1: about 46.1 ha of possible conversion (46.0% of the 1 km² cell) in the north-western part of the AOI (9.2 km from centre) [31.4627°N, 74.1864°E]. Hotspot #2: about 35.8 ha of possible conversion (35.8% of the 1 km² cell) in the north-western part of the AOI (8.5 km from centre) [31.4537°N, 74.1865°E]. Hotspot #3: about 34.3 ha of possible conversion (34.3% of the 1 km² cell) in the north-western part of the AOI (8.6 km from centre) [31.4628°N, 74.1969°E]. The top 3 one-square-kilometre cells hold 9% of all detected conversion, so the change is spread across many locations.

3. Can the results be visualized on a map?

Yes: static maps (RGB, NDVI, NDBI, land cover, change, hotspots), an interactive Folium map with layer toggles, and a Streamlit dashboard. Conversion polygons are exported as GeoJSON for QGIS / Google Earth.

4. Can the project be extended for district-level monitoring?

Yes. The AOI is one setting in the config: set aoi_mode='district' and district_name (e.g. 'Lahore', 'Kasur') and the same pipeline runs on the official FAO GAUL district polygon. Downloads are tiled, so larger areas work without code changes. Running it for several districts in a loop allows cross-district comparison.

10. Policy Recommendations (decision support)

- Use the hotspot list as a screening tool: send field teams or high-resolution imagery reviews to the top-ranked cells first, instead of surveying the whole area.
- Repeat the analysis every year in the same season to build an early-warning record of farmland loss; one model run for a district takes minutes once set up.
- Overlay the detected change with approved master-plan / land-use zoning and approved scheme boundaries. Only that comparison, not this model, can show whether a change was planned or unplanned.
- Combine vegetation loss with built-up gain (the urban heat-risk proxy) to prioritise green-space and tree-planting interventions in fast-growing cells.
- Treat all numbers as estimates with the stated uncertainty. They support, but do not replace, official surveys and legal processes.

11. Uncertainty, Assumptions and Data Sources

- Learning prototype - not an official government monitoring or enforcement system.
- Terms used: 'possible conversion', 'detected change', 'estimated change'. No claim is made about the legality or approval status of any development, housing society or landowner.
- Sentinel-2 resolution is 10-20 m. Small buildings, narrow roads and mixed pixels at the urban edge cannot be resolved reliably.
- A single season per year is compared. A field that is bare in one season can look like construction (false conversion), and active construction sites may still be classed as 'Other' (missed change).
- Accuracy depends on image quality, date selection, cloud cover, thresholds and validation quality. 2.92 km² of the detected conversion has low model confidence.
- Validation is based on visual interpretation of satellite imagery, which is itself uncertain for mixed or ambiguous points.
- Training-sample candidates came from global land-cover products (WorldCover, Dynamic World) and were filtered and checked visually. Errors in those products may still affect a few samples.

Data sources

- Copernicus Sentinel-2 MSI Level-2A surface reflectance, harmonized (ESA / Copernicus, via Google Earth Engine: COPERNICUS/S2_SR_HARMONIZED)
- Google Cloud Score+ for Sentinel-2 (GOOGLE/CLOUD_SCORE_PLUS/V1/S2_HARMONIZED) - cloud masking
- Google Dynamic World V1 (GOOGLE/DYNAMICWORLD/V1) - reference hints for sample selection
- ESA WorldCover 10 m 2021 v200 (ESA/WorldCover/v200) - reference hints for sample selection
- FAO GAUL 2015 administrative boundaries (FAO/GAUL/2015) - province/district outlines

12. LinkedIn / Portfolio Section

The full LinkedIn-ready write-up (all ten template sections, including the post text) is in **report/LINKEDIN_PORTFOLIO.md** and the repository README. Summary:

- **Project:** Punjab GeoGuardian - Detecting Urban Expansion Using Satellite Imagery (Lahore peri-urban expansion zone near Raiwind Road, 2019 vs 2025).
- **Tools:** Python, Google Colab, Google Earth Engine, Sentinel-2, Geemap, Folium, Scikit-learn (Random Forest), XGBoost, Rasterio, Streamlit, ReportLab.
- **Result:** about 12.31 km² of possible agricultural-to-urban conversion (5.47% of the AOI).
- **Validation:** 60 points, approximate accuracy 48.3%.
- **Deliverables:** Colab notebook, Python package, maps, interactive map, Streamlit dashboard, GeoJSON/CSV exports and this report.

Reproducibility

Configuration: random seed 42; season 02-01 to 03-31; cloud < 20%; 200 samples per class. Re-running notebooks/Punjab_GeoGuardian.ipynb reproduces every number in this report.